Vamadev Sundar Chebiyyam

bt24btech11024@iith.ac.in | c.vamadev@gmail.com | +91 9182996545 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.73
Rahul International School, Mira Road, Mumbai	Class XII (CBSE)	2024	93.40%
Gokuldham High School & Junior College, Goregaon (East), Mumbai	Class X (ICSE)	2022	98.20%

SCHOLASTIC ACHIEVEMENTS

- Consistently achieved A/A+ grades across the majority of biotechnology and bioinformatics coursework.

- **CGPA:** 8.73/10 -- B.Tech Biotechnology & Bioinformatics, IIT Hyderabad
- **SGPA**: 9.40/10 (latest semester)

POSITIONS OF RESPONSIBILITY

PR & Outreach Head, Biotechnology Society, IIT Hyderabad (2026-27):

In this capacity, I will be heading a team that focuses on:

1. **Public Relations**

- Collecting data on notable student and faculty achievements throughout the year.
- Gathering feedback for every society initiative.

2. **Outreach**

- Building and maintaining an alumni database in coordination with the department office.
- Organizing talks by alumni and eminent professionals (scientists, HR leaders, etc.) for career guidance -- covering topics such as:
- Securing a job or PhD abroad
- Preparing for CSIR / GATE / IELTS
- Crafting an effective resume for industry or academia

PR & Outreach Coordinator, Biotechnology Society, IIT Hyderabad (Aug 2025 - Jun 2026):

As PR&O Coordinator, I was responsible for contacting and coordinating with a number of highly noted industrial and academic individuals in the field of Biotechnology, both national and international, and also inviting them to various events, seminars, workshops, guest lectures, etc. conducted by the Biotechnology Society at IIT Hyderabad.

I was also involved in event planning, logistics and miscellaneous activities necessary for the smooth and successful function of the Society throughout the academic year.

Principal Undergraduate Researcher, **Project 6, Computational Systems Biology Lab, IIT Madras**:

Reporting directly to Prof. Karthik Raman. Tasked with developing an end-to-end computational pipeline for the rational design of microbial consortia and intelligent identification of drug targets for pathogen decolonization.

Computational Biologist, Team IITH, iGEM 2027:

Part of a select team tasked with building a synthetic-biology platform that computationally designs and biologically produces programmable, rapidly self-assembling protein materials that seal wounds on contact and actively guide tissue regeneration.

SKILLS

Genome-Scale Metabolic & Systems Biology Modeling:

Proficient in:

- COBRA Toolbox (MATLAB), COBRApy (Python)
- Flux Balance Analysis and its variants
- Flux Variability Analysis
- Model QC (MEMOTE)
- Energy-Generating-Cycle diagnosis
- Essential-reaction/synthetic-lethality screening (Fast-SL)
- Gap-filling (FASTCC)
- Shadow price/sensitivity analysis
- Gut-physiological medium modeling

Microbial Community & Ecological Modeling:

Experienced in:

- MICOM (joint-LP & competitive-fraction community FBA)
- Dynamic FBA (Monod/Michaelis-Menten kinetics)
- Cross-feeding analysis, minimal-consortium design (greedy set-cover)
- Reverse ecology/seed-set analysis, pathway enumeration (MetQuest)

Programming & Computational Methods:

Proficient in:

- Python (pandas, NumPy, NetworkX, COBRApy)
- MATLAB
- R (RevEcoR, plotting/visualization/statistical analysis)
- C
- Reproducible pipeline design
- Cross-toolbox validation and debugging

Working knowledge of:

- JavaScript, HTML, CSS
- React
- DBMS (relational database design & querying)

Applied Data Science:

As part of both my ongoing Research Projects, I am using:

- Bayesian optimization (Ax, BoTorch, BoFire)
- Design of experiments (DOE)
- Active/transfer learning concepts
- Research-data architecture (ELNs, model/version tracking)

Microbiology & Infection Biology:

My work on Rational Design of Microbial Consortia requires deep understanding of:

- Bacterial metabolism
- Antibiotic resistance mechanisms
- Gut colonization resistance
- Multi Drug Resistant (MDR) epidemiology
- Structured synthesis of primary literature

Research Strategy & Communication:

Research projects cannot be successfully executed without understanding how to build feasible strategies for research and being skilled at communication with fellow project members.

Biochemical and Bioprocess Engineering:

- Cell growth kinetics and batch/continuous stirred-tank fermentation modeling
- Kinetics of substrate utilization and product formation, enzyme kinetics and inhibition analysis (competitive, non-competitive, uncompetitive)
- Reaction thermodynamics (Gibbs free energy, enthalpy, entropy, chemical equilibrium)
- Bioprocess stoichiometry and mass/energy balances
- Transport phenomena (mixing, heat and mass transfer)

- Fermentation technology (submerged and solid-state, industrial fermenter design)
- Bioseparation and downstream processing (cell disruption, centrifugation, filtration, extraction, crystallization)
- Membrane processes (dialysis, ultrafiltration, microfiltration, reverse osmosis)
- Process control
- Scale-up and economic analysis of bioprocesses

RELEVANT COURSES

Course : Grade

Introduction to Life Sciences: A

Basic Bioinformatics: B

Biostatistics: B

Molecular & Cellular Biology: A

Next-Generation Sequencing: A

Protein Structure, Function &

Disease: A

Nucleic Acids and RNA Biomedicine: A

Molecular Biophysics and

Macromolecular Structural Biology: A

Differential Equations: A

Artificial Intelligence: A-

Biochemistry: A

Microbiology: A

Big Data Biology and

Biological Databases: A

Biological Engineering: A

R and Python for Biologists: A-

Basic Biotechnology Lab: A-

Biochemical Engineering: B

PROJECTS

Rational Design of Microbial Consortia for Pathogen Decolonization (Completed) (May - August 2026)
Computational Systems Biology Lab, IIT Madras -- under Prof. Karthik Raman | Undergraduate Researcher

This project applies genome-scale metabolic modeling (GEM) and constraint-based analysis, in combination with game-theoretic models and optimization frameworks to rationally design minimal microbial consortia for pathogen (specifically, Multi-Drug Resistant (MDR) pathogens) decolonization. As part of this project, I:

- Curated and built model/data inputs -- recipient taxonomy tables and genome selections -- for genome-scale community metabolic models of candidate decolonization consortia
- Designed a conditional variant of community FBA, which imposes rational constraints on the feasible flux polytope, and utilizes sampling methods to extract meaningful flux vectors for community members, to accurately simulate resource competition between consortia and pathogens, and eventually predict pathogen suppression
- Developed modified FBA formulations/algorithms incorporating game-theoretic dynamics (Stackelberg-style simulation and General Lotto framework of simultaneous competition) to better simulate resource competition between candidate consortia and the target pathogen, extending standard steady-state community FBA
- Conducted a literature review of gut colonization resistance and MDR clinical epidemiology to help define selection criteria for candidate consortium members

Project PHOENIX -- Synthetic-Biology Platform for Programmable, Self-Assembling Wound Sealing Protein Materials (Ongoing)

iGEM Team IITH -- under Prof. Gunjan Mehta | Computational Biologist

Project PHOENIX is a synthetic-biology platform that computationally designs and biologically produces programmable, rapidly self-assembling protein materials that seal wounds on contact and actively guide tissue regeneration and prevent infection.

Virtual Cell Challenge 2026 (Arc Institute) (Ongoing)

Team IIT Hyderabad | Team Member Arc Institute's

Virtual Cell Challenge 2026 benchmarks models predicting cellular responses to genetic/chemical perturbations. The 2026 task: zero-shot prediction of gene-knockdown responses across multiple cell lines from control profiles alone, scored on unseen experimental data.

Somatic & Germline Variant-Calling Pipeline (WGS/WES)

- Built and executed a complete variant-calling pipeline from raw paired-end FASTQ reads to annotated VCFs: quality control (FastQC), reference-genome indexing and alignment (BWA-MEM against hg38/hs37d5), and SAM-to-BAM conversion, sorting, and indexing (SAMtools)

- Assigned GATK read groups and built reference dictionaries, then ran tumor-normal somatic calling (Mutect2) and germline calling (HaplotypeCaller)

- Filtered and analyzed resulting VCFs with bcftools and visualized read alignments and called variants in IGV

Bulk RNA-Seq Differential Expression Pipeline

- Ran an end-to-end bulk RNA-seq pipeline from raw FASTQ reads: QC/trimming (FastQC, Trimmomatic/fastp), alignment (HISAT2), gene-level quantification (featureCounts)

- Performed differential expression analysis (DESeq2) to identify differentially expressed genes, with exploratory and results visualization via PCA and volcano plot